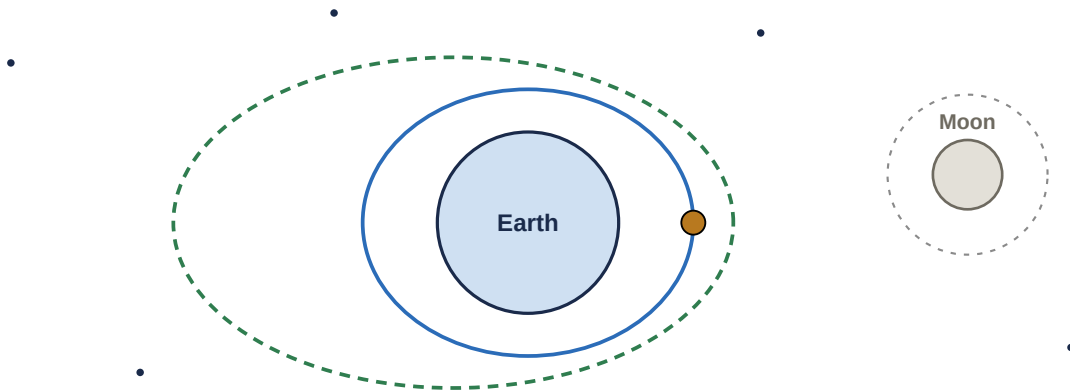


My Orbit Workbook

How to fly a spaceship around the Earth — and all the way to the Moon!



This workbook belongs to:

Works together with **Dan's Orbit Simulator**
goodorigamiman.github.io/dans-orbit-simulator

How to use this workbook

This book goes with the **Orbit Simulator** — the computer game where you fly a little spaceship around the Earth. The simulator has 9 lessons. This book has the same 9 lessons.

For each lesson: **1)** read the Big Idea here, **2)** try it in the simulator, **3)** come back and finish the puzzles on paper. Answers are hiding on the last page — no peeking until you try!

Space words (your astronaut dictionary)

orbit	the path a spaceship follows as it circles a planet or moon
prograde	firing your engine FORWARD, the way you are already going (speeds you up)
retrograde	firing your engine BACKWARD, against your motion (slows you down)
high point	the tallest spot of your orbit (grown-ups say "apoapsis")
low point	the lowest spot of your orbit (grown-ups say "periapsis")
fuel push (delta-v)	how much total push your engine has given — the fuel money of space
escape velocity	the speed where you break free and never come back
sphere of influence	the Moon's neighborhood — inside it, the Moon's gravity is the boss
transfer	a trip from one orbit to another
capture	getting caught by a moon's gravity so you loop around it

LESSON 1

What is an orbit?

The Big Idea

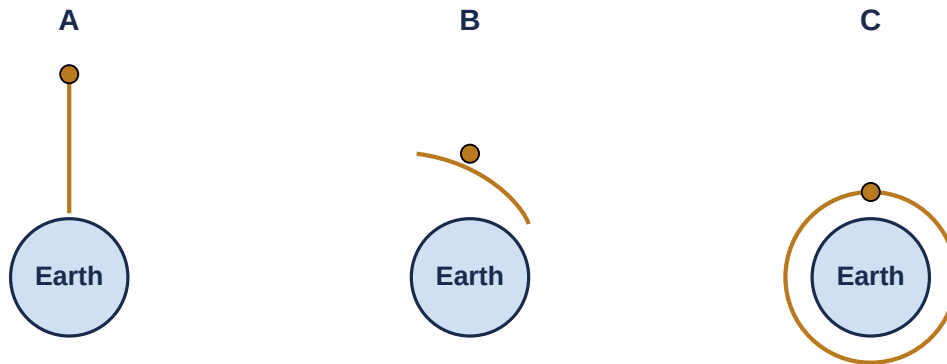
A spaceship in orbit is **falling** the whole time! It moves sideways SO fast that as it falls, the round Earth curves away underneath it. So it keeps falling... and keeps missing the ground... forever. That is an orbit: **falling sideways and always missing.**

Try it in the simulator

Open Lesson 1. Just watch the ship go around once or twice. Slide the **time warp** to make time go faster. Nothing is holding the ship up!

Your turn

1. A ball is let go near Earth in three different ways. **Circle the picture** that shows an orbit:



2. Fill in the blank: An orbit is falling _____ forever.
3. True or false (circle one): Astronauts float because there is no gravity in space.
TRUE / FALSE

Hint: they float because they are FALLING, together with their ship!

LESSON 2

Raise the far side

The Big Idea

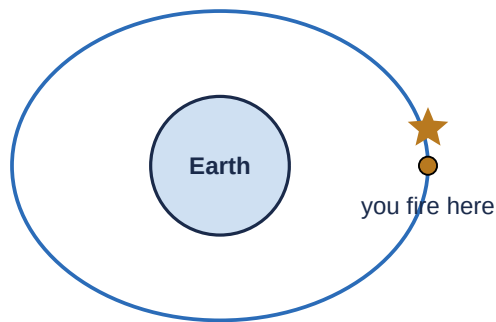
Here is the most surprising rule in all of space flying: when you fire your engine, your orbit does NOT change where you are. It changes on the **OPPOSITE side** of the planet — half a lap away! Speeding up makes the **far side** of your loop stretch up higher.

Try it in the simulator

Open Lesson 2. Hold the **Prograde (forward)** button and watch the orange "changes here" marker — it is always across from you! Push your high point above 1,000 km.

Your turn

1. The ship fires its engine at the gold star. **Draw a big X** where the orbit will get higher:



2. Fill in the blank: A burn changes the _____ side of the orbit.

3. Copy the rule (this one is worth remembering forever):

"Where I burn decides how much. Where the orbit changes is across from me."

LESSON 3

Where you burn matters

The Big Idea

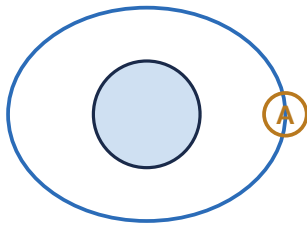
Same engine. Same push. Different spot = a totally different orbit! If you burn at point A, the bulge grows across from A. If you burn at point B, the bulge grows across from B. The **place** you burn picks the **shape** of your new orbit.

Try it in the simulator

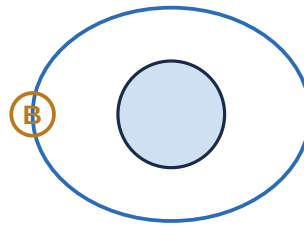
Open Lesson 3. Burn 40 m/s near marker A, look at the shape. Then reset and burn the same amount near B. Compare!

Your turn

1. In each picture, **draw the bulge** (the stretched part of the orbit) where it belongs:



Picture 1: burn at A



Picture 2: burn at B

2. Predict, then check in the simulator: If you burn at the TOP of the orbit, the bulge appears at the _____ of the orbit.
3. Circle one: The engine decides **how much** the orbit changes. The burn spot decides **WHERE / WHY** it changes.

LESSON 4

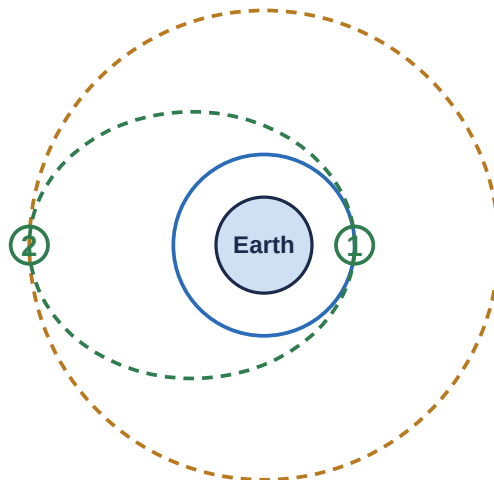
The two-burn transfer

The Big Idea

One burn can only stretch ONE side of your orbit. So how do you move your WHOLE orbit higher? You need **two burns**: **Burn 1** stretches the far side up. Then you coast to the top and do **Burn 2**, which lifts the side you left behind. Two burns, two places — and your orbit is a big circle. Real rocket scientists call this a **Hohmann transfer**.

Try it in the simulator

Open Lesson 4. Use the **flight plan**: add burn 1 at the low point and burn 2 at the high point, look at the dashed preview, then press Execute and watch both burns fly.



Burn 1 stretches the orbit up. Burn 2 makes it round again.

Your turn

1. Put these steps in order. Write 1, 2, 3, 4:

- ___ Coast (no engine) halfway around, up to the new high point
- ___ Fire forward to stretch the far side up
- ___ Enjoy your new, bigger circle!
- ___ Fire forward again to lift the low side

2. Fill in the blank: Moving BOTH sides of an orbit takes _____ burns.

LESSON 5

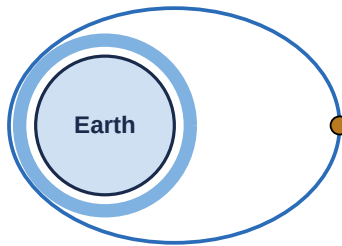
Coming home

The Big Idea

To come home, you do NOT point at the Earth and fire. You simply **slow down**. Braking lowers the opposite side of your orbit until it dips into the **air blanket** around Earth (the atmosphere). Then the air rubs against your ship and slows you down the rest of the way — for free!

Try it in the simulator

Open Lesson 5. Hold **Retrograde (backward)** until your low point drops under 80 km. Then keep coasting and watch re-entry happen.



The low point dips into the air blanket - the air slows you down for free.

Your turn

1. Circle the right way to come home:

- a) Point at the Earth and fire the engine hard
- b) Slow down, and let the orbit dip into the air

2. Fill in the blank: The _____ slows the ship down for free at the end.

3. True or false: You never "fly down" — you slow down HERE and the orbit dips over THERE. **TRUE / FALSE**

LESSON 6

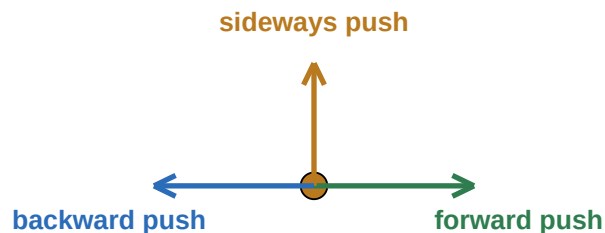
Steering the burn

The Big Idea

Engines can point in different directions! A **forward** push makes your orbit BIGGER. A **backward** push makes it SMALLER. A **sideways** push mostly just **tilts and turns** the orbit — it hardly grows at all. That is why real astronauts almost always burn forward or backward: it buys the most orbit for your fuel.

Try it in the simulator

Open Lesson 6. Slide **Steer** to 60° or more, then hold Prograde and burn 40 m/s. See how little the orbit grows? Now try the same burn with Steer at 0°.



Your turn

1. Draw a line to match each push to what it does:

- | | | | |
|---------------|---|---|-------------------------|
| forward push | • | • | tilts / turns the orbit |
| sideways push | • | • | makes the orbit bigger |
| backward push | • | • | makes the orbit smaller |

2. Circle one: To grow your orbit with the least fuel, push **FORWARD / SIDEWAYS**.

LESSON 7

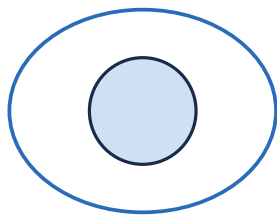
Breaking free (escape velocity)

The Big Idea

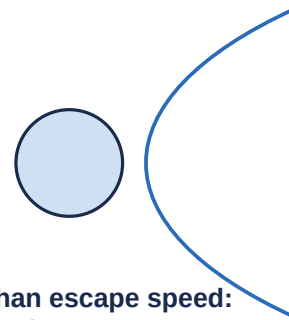
At every height there is a magic speed called **escape velocity**. Go slower, and gravity always bends your path into a loop that comes back. Go faster, and the loop **tears open** — you fly away and NEVER come back. Near Earth, circling speed is about 7.7 km every second. Escape speed is about 10.9 — about **1.4 times** faster. That "1.4 times" rule works at every planet and moon!

Try it in the simulator

Open Lesson 7. Execute the big burn (or hold Prograde a long time) and watch the "high point" number climb... and climb... until it says **infinity** (a sideways 8). Your path is open — you broke free!



slower: path stays closed



faster than escape speed:
the path tears open!

Your turn

1. Near Earth: 7.7 km/s = circling speed. 10.9 km/s = escape speed. A ship is going 9 km/s. Circle one: it **STAYS / LEAVES**.
2. Fill in the blank: When you break free, the high point reads _____ (a sideways 8!).
3. True or false: Once past escape speed you can still turn back if you brake soon.
TRUE / FALSE

LESSON 8

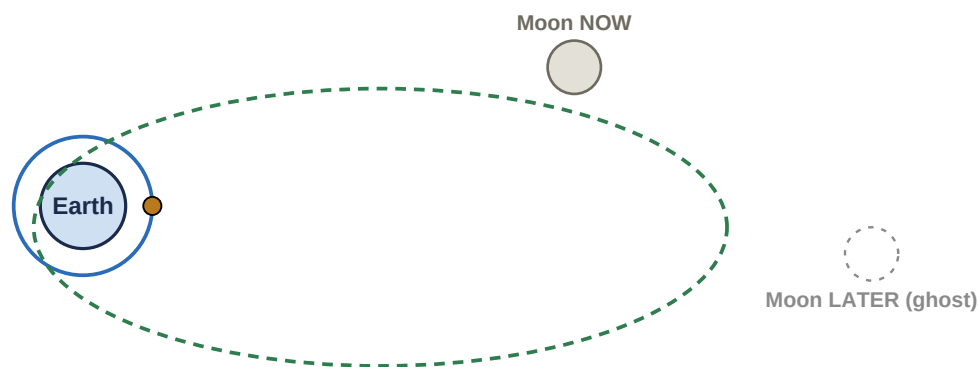
Lead the Moon

The Big Idea

The Moon is always **moving**. If you aim at where the Moon is NOW, you will arrive at empty space — the Moon will have moved on! You must throw yourself at where the Moon **WILL BE**, like leading a running friend with a football. Space pilots call the right moment to leave a **transfer window**.

Try it in the simulator

Open Lesson 8. Mission control loaded your departure burn already. See the **ghost Moon**? That is where the Moon will be when you arrive. Execute the plan and coast into the Moon's neighborhood (the dashed circle).



Your turn

1. You throw a ball to a friend who is running. Where do you throw it? Circle one:
a) Right at your friend b) At the spot your friend is running TO
2. Fill in the blank: The dashed circle around the Moon is its sphere of _____ — inside it, the Moon's gravity is the boss.
3. In the picture above, **draw an arrow** from the rocket's path to the ghost Moon — that's your meeting spot!

LESSON 9

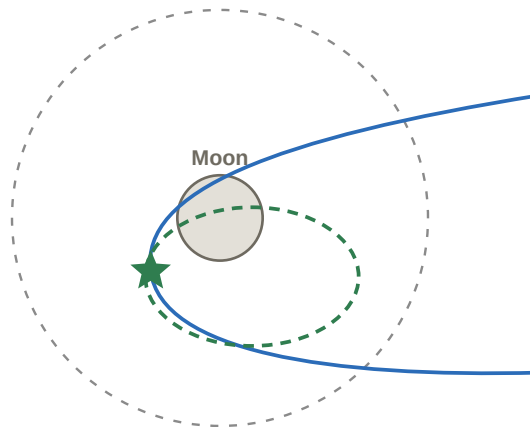
Catching the Moon (lunar orbit insertion)

The Big Idea

You arrive at the Moon going **TOO FAST** to stay. Do nothing, and the Moon's gravity slings you right back out into space! **Arriving is braking:** at your closest point to the Moon, fire **backward** until your path closes into a loop. Now the Moon's gravity can hold you — you are **captured**. Every real Moon mission does exactly this. And careful: the Moon has **no air**, so nothing slows you down for free there!

Try it in the simulator

Open Lesson 9. Mission control loaded the braking burn. Execute it — or try flying past **WITHOUT** braking first, and feel the slingshot throw you home!



Brake at the star (closest point) and the path closes into a loop!

Your turn

1. Fill in the blank: To get captured, brake when you are _____ to the Moon.
2. Circle one: On the Moon there is **NO AIR / THICK AIR**, so all your braking must come from your engine.
3. True or false: If you skip the brake, the Moon slings you back toward Earth.
TRUE / FALSE

The Big Astronaut Quiz

You made it through all 9 lessons! Show what you know. (Answers on the last page.)

Part 1 — Match the word to its meaning (draw lines)

- | | | | |
|-----------------|---|---|--------------------------------------|
| prograde | • | • | the lowest spot of an orbit |
| retrograde | • | • | firing your engine forward |
| high point | • | • | getting caught by the Moon's gravity |
| low point | • | • | the speed of no return |
| escape velocity | • | • | firing your engine backward |
| capture | • | • | the tallest spot of an orbit |

Part 2 — True or false (circle)

1. A burn changes the orbit on the opposite side. **T / F**
2. One burn is enough to make your whole orbit bigger and round. **T / F**
3. To come home from Earth orbit, you slow down. **T / F**
4. To aim for the Moon, fire toward where the Moon is right now. **T / F**
5. Sideways pushes are the best way to make an orbit bigger. **T / F**
6. At the Moon, you must brake with your engine — there is no air to help. **T / F**

Certificate of Orbit Mastery

This certifies that space pilot

has learned how orbits really work:

burns change the **opposite side** · big moves take **two burns** ·
coming home means **slowing down** · breaking free takes **escape velocity** ·
you aim where the Moon **will be** · and **arriving is braking**.

Teacher / Mission Control

Date

Answer key (no peeking until you tried!)

Lesson 1	1: picture C. 2: "sideways". 3: FALSE — they float because they are falling with their ship.
Lesson 2	1: the X goes on the LEFT side, straight across from the star. 2: "opposite".
Lesson 3	2: bottom. 3: WHERE.
Lesson 4	1: order is 2, 1, 4, 3. 2: two.
Lesson 5	1: b. 2: "air" (the atmosphere). 3: TRUE.
Lesson 6	1: forward = bigger, sideways = tilts/turns, backward = smaller. 2: FORWARD.
Lesson 7	1: STAYS (9 is less than 10.9). 2: infinity (the sideways 8). 3: TRUE.
Lesson 8	1: b. 2: "influence".
Lesson 9	1: closest. 2: NO AIR. 3: TRUE.
Quiz Part 1	prograde = forward · retrograde = backward · high point = tallest spot · low point = lowest spot · escape velocity = speed of no return · capture = caught by the Moon.
Quiz Part 2	1 T · 2 F (you need two burns) · 3 T · 4 F (aim where it WILL be) · 5 F (forward pushes grow orbits best) · 6 T.

Made to go with Dan's Orbit Simulator — goodorigamiman.github.io/dans-orbit-simulator